

Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

The **Rising Waters – Intelligent Flood Prediction and Early Warning System** is designed using a modular architecture with Python, Flask, and a Decision Tree Machine Learning model. The current system works efficiently for local deployment and can be enhanced to support larger datasets, more users, cloud deployment, and real-time weather data integration.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical dataset only	Cannot provide real-time flood prediction	High
2	Local deployment	Limited accessibility and scalability	Medium
3	No automated alert system	Users must manually check predictions	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on cloud platforms (AWS, IBM Cloud, Azure) to support many users
2	Data Storage	Local CSV dataset	Use cloud databases such as MySQL, PostgreSQL, or IBM Cloudant
3	Performance	Decision Tree model on local machine	Optimize prediction pipeline and use cloud servers for faster response
4	Security	Basic input validation	Add user authentication, HTTPS, encryption, and secure APIs

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Deploy the application to the cloud	1 Month	Improve accessibility and availability	Deploy the application to the cloud
Integrate real-time Weather API	2 Months	Provide live and more accurate flood predictions	Integrate real-time Weather API
Add SMS and Email alert notifications	3 Months	Notify users instantly during flood risk	Add SMS and Email alert notifications